

Skills Summary

Vertex, Axis, Intercepts, and Opening Direction

Use this reference while completing your worksheet or when studying.

Skill 1 — The axis of symmetry and the vertex

For $f(x) = ax^2 + bx + c$: the axis of symmetry is the vertical line $x = -\frac{b}{2a}$.

The vertex sits on that line, so its first coordinate is that value of x and its second coordinate is f evaluated there. Find the line first, then substitute — never the other way round.

Sign care: b carries its own sign into the formula, so a negative b gives a positive axis.

Example: Find the axis and the vertex of $f(x) = x^2 - 8x + 11$.

Step 1. The vertex is the one point on the axis of symmetry, so locating the axis is what makes the vertex a single substitution.

Step 2. $a = 1$, $b = -8$, so the axis is $x = -\frac{-8}{2(1)} = \frac{8}{2}$.

Step 3. Substitute: $4^2 - 8(4) + 11 = 16 - 32 + 11$.

axis $x = 4$, vertex y -value $= -5$

Try it: Find the axis and the vertex y -value of $f(x) = x^2 + 6x + 2$.

check: axis $x = -3$, vertex $y = -7$

Skill 2 — Both kinds of intercept

y -intercept: the output at input 0, which for $ax^2 + bx + c$ is simply c .

x -intercepts: the inputs that make the output 0. Factor and use the zero-product property — a product is zero only when one of its factors is.

The sign flip: a factor $(x + 6)$ gives the intercept $x = -6$, not 6. Solve each bracket rather than copying its number.

Example: Find both kinds of intercept for $f(x) = x^2 - x - 6$.

Step 1. Setting the output to zero is what the x -intercepts mean, and setting the input to zero is what the y -intercept means — two different substitutions.

Step 2. $x^2 - x - 6 = (x + 2)(x - 3)$, so each bracket is zero at $x = -2$ and $x = 3$.

Step 3. At input 0 the value is the constant term, -6 .

x -intercepts -2 and 3 ; y -intercept -6

Try it: Find both kinds of intercept for $f(x) = x^2 - 2x - 15$.

check: y -intercept -15 and x -intercepts 5 and -3

Skill 3 — Which way it opens, and what that means

Rule: $a > 0$ opens *upward*, so the vertex is the **lowest** point — a minimum. $a < 0$ opens *downward*, so the vertex is the **highest** point — a maximum.

Why: the squared term dominates for inputs far from the vertex, and its sign decides whether those far-away outputs run up or down.

Use it as a check: a downward parabola whose vertex you computed as a low point means a slip somewhere.

Example: For $f(x) = -2x^2 + 4x + 1$, find the vertex and say what kind of point it is.

Step 1. The leading coefficient is negative, so the curve opens downward and the vertex will be the maximum — worth knowing before computing it.

Step 2. Axis: $x = -\frac{4}{2(-2)} = -\frac{4}{-4}$.

Step 3. Value there: $-2(1)^2 + 4(1) + 1 = -2 + 4 + 1$.

axis $x = 1$, maximum value **3**

Try it: For $f(x) = -x^2 + 6x - 5$, find the axis and the maximum value.

check: axis $x = 3$, maximum **4**